

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Placebo-Controlled, Double-Blinded Trial to Evaluate the Safety, Tolerability, and Immunogenicity of a Multivalent Group B Streptococcus Vaccine in Healthy Pregnant Women and Their Infants **Short Title/Acronym:** BEATRIX: A Study to Learn About a Group B Streptococcus Vaccine in Healthy Pregnant Women and Their Babies **Protocol Number:** C1091009 **NCT Number:** NCT07160244 **Posting ID:** 1108998779051164767 **Trial Status:** RECRUITING **Sponsor/Company Name:** Pfizer Inc. **Investigational Product:** Multivalent Group B streptococcus vaccine (GBS6, 6-valent polysaccharide conjugate vaccine) **Preferred UMLS Name:** Streptococcus agalactiae Vaccines **Phase of Development:** Phase III **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and immunogenicity of a multivalent Group B Streptococcus vaccine (GBS6) in healthy pregnant women, and to assess the protective levels of passively transferred maternal anti-capsular polysaccharide (CPS) antibodies in their infants at birth. **Secondary Objectives:** To evaluate the infant's serotype-specific geometric mean ratio (GMR) of antibodies at birth and the maternal GBS serotype-specific IgG levels at 1 month post-vaccination and at delivery.

2.2 Background & Rationale Group B streptococcus (GBS) is a major cause of neonatal disease, including early-onset and late-onset disease. Natural history studies correlate maternally transferred anti-CPS antibodies with protection against infant GBS disease. A maternal GBS six-valent polysaccharide conjugate vaccine (GBS6) is being developed to protect neonates and infants up to 3 months of age through the passive transfer of antibodies from the mother to the infant.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (saline) **Blinding:** Double-blind **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Target \$N\$: 6,000 estimated pregnant participants **Disease Area:** Healthy **Number of Sites:** Approximately 105 global locations **Estimated Study Start:** 25-Aug-2025 **Estimated Primary Completion:** 06-Nov-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy pregnant women ≤ 49 years of age who are between 24 0/7 and 36 0/7 weeks of gestation on the day of planned vaccination. 2. Uncomplicated, singleton pregnancy with no known increased risk of complications, and a fetal anomaly ultrasound examination with no significant fetal abnormalities observed. 3. Documented negative human immunodeficiency virus (HIV) antibody test, syphilis test, and hepatitis B virus (HBV) surface antigen test during this pregnancy and prior to randomization. 4. Capable of giving personal signed informed consent and willing to give informed consent for her infant to participate in the study. 5. **Infant Participants:** Evidence of a signed and dated Informed Consent Document (ICD) signed by the parent(s)/legally authorized representative or legal guardian.

4.2 Exclusion Criteria 1. Prepregnancy body mass index (BMI) of > 40 kg/m². 2. Current or prior pregnancy complications or abnormalities that may increase the risk associated with the participation in and completion of the study. 3. History of microbiologically proven invasive disease caused by GBS in the current pregnancy, or a known/suspected infection during the current pregnancy that may increase the risk of complications (e.g., active tuberculosis, syphilis, primary genital herpes simplex, malaria). 4. **Infant Participants:** Children or grandchildren who are direct descendants of investigator site staff or sponsor and sponsor delegate employees directly involved in the conduct of the study. 5. **Infant Immunogenicity Subset Participants:** Children with a known or suspected contraindication to any vaccine administered in the infant vaccine immunogenicity subset.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose (1 shot) **Route of Administration:** Intramuscular injection in an arm **Duration of Treatment:** Single administration. Follow-up for a maximum of 14 months (6 months after delivery) for mothers, and about 12 months for their babies (up to 19 months for a subset of infants).

5.2 Concomitant & Prohibited Therapy Permitted: Routine prenatal care medications. A subset of infants will receive standard-of-care immunizations (diphtheria toxoid-containing vaccine and/or pneumococcal vaccine, e.g., Infanrix, Prevnar) following their country's standard immunization plan. **Prohibited:** Concurrent investigational therapies or vaccines outside of the authorized standard-of-care windows.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Baseline	Day 0	Informed Consent, Medical History, Ultrasound Review, Randomization, Single Shot Vaccination
Interim	Up to Delivery	3 to 4 visits (including permitted telephone follow-ups), Safety E-diary reviews, Blood draw 1-month post-vaccination
Delivery	Day of Birth	Maternal & cord blood draws to assess antibody transfer, Infant baseline assessment
End of Study (Maternal)	6 Months Post-Delivery	Final safety check and exit interview
End of Study (Infant)	12 to 19 Months	Infant safety follow-up, blood draws 1 month after completion of standard primary and/or toddler (booster) immunizations

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. The proportion of maternal participants reporting prespecified local reactions within 7 days following study intervention. 2. The proportion of maternal participants reporting prespecified systemic events within 7 days following study intervention. 3. The proportion of maternal participants reporting adverse events (AEs) through 1 month following study intervention.

7.2 Secondary & Exploratory Endpoints 1. To measure GBS serotype-specific opsonophagocytic titers in infant participants at birth by geometric mean ratio (GMR) between the GBS6 group versus the placebo group, and by the difference in seroresponse rates between the GBS6 group and placebo group. 2. To describe anti-CPS IgG antibody levels predicted to provide protection from invasive GBS disease (all disease) caused by the 6 individual vaccine serotypes in infants when GBS6 is administered to healthy pregnant women. 3. To describe anti-CPS IgG antibody levels in infant participants born to maternal participants vaccinated with GBS6.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient diary for 7 days post-vaccination for local/systemic reactions. Long-term monitoring via scheduled clinic visits and telephone calls up to 1 month post-vaccination for general AEs. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for any SAEs affecting the mother or the infant. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee assigned to continuously evaluate maternal and neonatal safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Safety Set for vaccinated mothers and their live-born infants. **Sample Size**

Justification: \$N=6,000\$ appropriately powered to detect safety profiles and statistically significant differences in immune responses and maternal antibody transfer rates versus placebo.

Handling of Missing Data: Missing serology data handled via prespecified imputation algorithms; safety data analyzed as observed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure integrity of maternal and pediatric data. **Audit Trail:**

Validated eCRF systems, prompt data clarification forms, and secure, blinded handling of serology results.

11. Study Identifiers & Governance

Primary ID: C1091009 **Posting ID:** 1108998779051164767 **Registry Number:** NCT07160244 **Trial Status:** RECRUITING **Sponsor/Lead**

Organization: Pfizer Inc. **Study Title:** BEATRIX: A Study to Learn About a Group B Streptococcus Vaccine in Healthy Pregnant Women and Their Babies **Official Regulatory Title:** A Phase 3, Randomized, Placebo-Controlled, Double-Blinded Trial to Evaluate the Safety, Tolerability, and Immunogenicity of a Multivalent Group B Streptococcus Vaccine in Healthy Pregnant Women and Their Infants

12. Clinical Framework (Maternal Vaccine Specific)

Primary Condition: Group B Streptococcus (GBS) (Disease Area: Healthy) **Diagnosis Threshold:** Healthy pregnancies between 24 0/7 and 36 0/7 weeks of gestation **Medical Methodology:** Maternal Immunization **Optimization Target:** Passive transfer of protective anti-CPS antibodies to the neonate

13. Study Procedures & Methodology

Management Pathway: Standard antenatal care combined with a single investigational vaccination **Education Protocol (HCP):** Site initiation visits and training on vaccine administration/pediatric follow-up **Education Protocol (Patient):** Informed consent and training on using e-diaries for capturing post-vaccination reactions **Quality Audit Mechanism:** Centralized safety monitoring and standardized immunological assay processing

14. Observation & Follow-Up Schedule

Total Duration: Up to 14 months (maternal) / Up to 19 months (pediatric subset) **Onsite Visit Cadence:** 3 to 4 maternal visits (with some permitted phone contacts) **Remote Monitoring Frequency:**

Intermittent protocol-defined safety phone calls **Checklist Review Interval:** Prompt review of 7-day post-vaccination e-diaries

15. Sign-off & Approval

Role	Name	Signature	Date			:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]						